

1 THE TASK



CQ-DINO: Mitigating Gradient Dilution via Category Queries...
NeurIPS 2025 · arXiv 2503.18430



Match target

cheapest config that proves the paper’s claim

metric AP / AP50 / AP75
scope V3Det val · 29,821 imgs
target 46.3 / 51.5 / 48.4 ±5%



configuration withheld



What the authors released

- ✓ data
- ✓ code
- ✓ weights

AVAILABILITY TIER

► Run execute what shipped

Retrain weights withheld

Reimplement code withheld too

the availability cliff
success falls as the release thins

budget 8 H100-h

×100 papers

2 THE RUN



Reproduction agent

deepseek-v4

workspace_bash

write_file

fetch_url



run_gpu

```
run 2889575 · 107 rounds · 6.0 of 8 H100-h

PLAN · r0-4
● [plan] AP 46.3 on V3Det · design eval, run it

FETCH · r5-8
○ $ git clone github.com/.../CQ-DINO
○ $ gdown ...swinb1k_v3det.pth # 1.89 GB

BUILD · r9-61
○ $ pip install mmcv==2.2.0
● ERROR: no aarch64 wheel
○ AH! no aarch64 wheel. build it from source
○ $ FORCE_CUDA=1 pip install -e mmcv/

RUN · r62-64
● $ run_gpu tools/test.py # 29,821 imgs GPU
● srun: task 0: Out Of Memory

RESCUE & SCORE · r65-92
○ OOM again, but the preds survived in /tmp
○ $ cp /tmp/results.bbox.json evidence/
● $ run_gpu chunked_eval.py # score them GPU

VERIFY & REPORT · r93-107
○ my scorer agrees with COCOeval, twice
○ $ write_file report.json
```

bbox_mAP 0.463 0.515 0.484 = paper’s 46.3 / 51.5 / 48.4 ✓

● plan ○ thought ○ step ● GPU run ● error \$ agent command

3 THE REPORT

report.json **claims: "reproduced"**
evidence/ eval.log · commands.log
results.bbox.json · patches/
every number traces to a logged run

report + evidence

4 THE AUDIT



Adversarial audit

re-derives the verdict from the evidence

- ✓ metric recomputed from raw predictions
- ✓ exact checkpoint, sibling rows ruled out
- ✓ AP_{S/M/L} values not in the paper: no echo
- 🚩 training arm untested (released ckpt only)

THE RUN CLAIMS
AP 46.3 ✓

vs

THE AUDIT FINDS
AP 46.3 ✓

RECLAIM scores the audited verdict